

Causal but Not Carrier: Canonical Entropy Neurons Influence but Do Not Encode the Variation of Output Entropy

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June 11, 2026

Abstract

Entropy neurons are late-layer units that write predominantly to the null space of the unembedding matrix and modulate a model’s output entropy through the scale of the final LayerNorm, with little direct effect on the argmax token. Because they offer a clean, localized handle on predictive uncertainty, they are read as a confidence-regulation mechanism. We distinguish two properties that this reading conflates: whether these neurons *influence* entropy under intervention, and whether they *carry* the variation of entropy that a model exhibits on natural text. Across 13 models from 7 architecture families, the two dissociate. Output entropy is linearly decodable from the input to the final MLP in every model that has entropy neurons (held-out $R^2_{\text{full}} = 0.55\text{--}0.74$), and the canonical entropy neurons causally move it in every family that contains them (clip-ablation $|\Delta H|$ up to ~ 2.5 nats in OPT, whose random baseline is non-zero; the largest effect over a near-zero baseline is ~ 1.4 nats; intervals exclude zero). The same neurons linearly carry entropy variance only in the GPT-2 lineage ($R^2_{\text{EN}} = 0.21\text{--}0.40$; paired-bootstrap margin over a random-neuron baseline strictly positive and stable across seeds), and in no other family. The dissociation between a large causal effect and a carried variance indistinguishable from random is itself robust across the seven non-GPT-2 models with entropy neurons; that carrying appears *only* in the GPT-2 lineage is a separate and more tentative single-lineage claim. The absence is not an artifact of probe weakness: positive controls—probes from correlation-with-entropy neurons and from principal components—are high in all families but two (BLOOM and OPT, where the PCA control is degenerate), so the low R^2_{EN} outside the GPT-2 lineage reflects a genuine absence of linear role. The direction of the causal effect further dissociates from carrying: it is family-dependent, raising entropy in Pythia, GPT-Neo, and OPT and lowering it in GPT-2, BLOOM, and Qwen, so an identical mechanistic signature need not imply an identical role. Where the variance does reside is the residual *direction* rather than its *norm*—recoverable considerably more readily from the former, and distributed over ~ 256 principal directions. We release a sign-invariant attribution measure validated on synthetic data, show that the correlation-with-entropy and null-space criteria select disjoint neuron sets, and replace a naive gap statistic with a paired bootstrap. We reproduce the prior causal result and add the carrier measurement; we characterise the split but do not claim to fully explain it. We then ask directly whether some other residual direction carries entropy where the canonical neurons do not. Across nine further architectures we find exactly one: GPT-Neo-125M has a one-dimensional, entropy-neuron-orthogonal (write-subspace overlap 0.008), bidirectionally controllable causal carrier that beats a principal-component control at every tested rank—we are not aware of a prior report of such a carrier outside the GPT-2 lineage—yet the same distinct-carrier property is absent in, and the underlying direction is qualitatively different rather than merely weaker in, GPT-Neo-1.3B, an identical architecture and corpus at larger scale. A companion continued-pretraining experiment, run to isolate the pretraining-domain confound, finds that GPT-2’s carrying decays with continued training even on its own native domain,

most of it attributable to generic fine-tuning rather than the domain switch we manipulate, evidence against a clean data-alone account. Neither result explains why the GPT-2 lineage or GPT-Neo-125M specifically carry entropy; together they narrow the candidate explanation from architecture or corpus identity toward a scale- or capacity-dependent mechanism, and they extend a second, independent dissociation—between what a direction predicts and what intervening on it does—across architectures, complementing a concurrent, single-model observation of the same phenomenon in a different context [8].

1 Introduction

A language model’s calibration—how well its predictive entropy tracks its error rate—bears on hallucination, selective prediction, and downstream decisions. Mechanistic accounts of calibration have centered on *entropy neurons* (EN) [19, 10]: high-norm units in the last few layers whose output weights lie almost entirely in the null space of the unembedding, so that activating them changes the norm of the residual stream—and hence the scale of the final LayerNorm—without appreciably changing the top token. Intervening on them shifts output entropy, which has been read as evidence for a confidence-regulation function.

This inference is incomplete. A component can set a quantity under intervention while accounting for little of its variation across ordinary inputs. We therefore separate three quantities, formalized in Section 3.1: a neuron set’s causal *influence* on entropy under ablation; the share of natural entropy variation it linearly *carries* (used interchangeably with *encodes*); and the *decodability* of entropy from the full residual stream. Decodability does not imply encoding in any particular neuron set, and that gap is the substance of our result: across 13 models from 7 families, entropy is decodable everywhere (held-out $R^2_{\text{full}} = 0.55\text{--}0.74$) and the canonical entropy neurons influence it everywhere, yet they carry its variation only in the GPT-2 lineage.

Contributions.

1. *A carrier-versus-causal dissociation* (Figure 2, Table 1). Across 13 models in 7 families, canonical entropy neurons causally move output entropy wherever they exist but linearly carry its variance only in the GPT-2 lineage, established by a paired-bootstrap comparison against a random-neuron baseline that is stable across selection seeds. The dissociation itself holds across all seven non-GPT-2 models with entropy neurons; the further claim that carrying is confined to one lineage is stated more cautiously.
2. *A probe-capability control* that excludes the deflationary reading: positive controls recover entropy in exactly the families where the entropy-neuron probe is at chance, so the absence is specific to these neurons.
3. *A mechanistic characterisation* (on four anchor models): entropy is decodable from the residual direction considerably more than from its norm, and is distributed over ~ 256 principal directions rather than concentrated in one. We additionally exclude the near-constant-activation explanation by showing that carrying is inversely related to entropy-neuron activation variance.
4. *Methodological tools and corrections*: a sign-invariant attribution measure validated on synthetic data; the observation that the correlation-with-entropy and null-space criteria select disjoint neuron sets; and a corrected gap statistic that replaces a difference of independent marginal intervals with a paired bootstrap.

5. *Scope*. The carrying set is the GPT-2 family—four models, one a distillation of another—along which architecture, pre-training data, and evaluation-domain match are confounded. We report the split as distinguishable, not explained.
6. *A distributed carrier outside the entropy neurons, and its scale-dependence* (Section 4.7). Across nine further architectures under an interchange-transfer and causal-necessity protocol matched against random, PCA, and temperature controls, GPT-Neo-125M has a clean, EN-orthogonal, bidirectionally controllable causal carrier that beats PCA at every tested rank; it is the only one of the nine. GPT-Neo-1.3B, identical in architecture and corpus, does not reproduce it, and the underlying direction is qualitatively rather than quantitatively different (a sign-flipped partial correlation with output entropy), which we use in Section 4.8 to rule out several single-axis architectural explanations.
7. *Prediction is not intervention, across architectures* (Section 4.7). A ridge probe recovers entropy from the same residual site at $R^2 \approx 0.51\text{--}0.75$ already at rank one in every model tested, while low-rank causal controllability exists only where the interchange test passes—the dissociation holds across architectures, complementing an independent, single-model observation of the same underlying phenomenon reported concurrently in a different context [8].
8. *Partial evidence on the domain confound* (Section 4.9). A single-seed continued-pretraining pilot aimed at the evaluation-domain confound of Section 4.4 finds that GPT-2’s carrying decays with continued training even on its own native domain, most of it attributable to generic fine-tuning; this is evidence against a clean domain-alone account, though it bears on decay under continued training rather than origin at pretraining time, and awaits the multi-seed confirmatory run its own pre-registration calls for.

2 Related Work

Entropy neurons were first noted as a universal functional family in GPT-2 models [10] and then characterized as units writing to the unembedding null space and regulating entropy through the final LayerNorm, with causal interventions demonstrated across models up to several billion parameters [19]. That work establishes a *causal* role and observes that the effect varies across models, but it does not claim that the neurons carry the variation of entropy on natural text. We take this complementary measurement: we reproduce the causal effect (intervals exclude zero in every model with entropy neurons) and add a carrier test over a wider model set, pairing each causal claim with a probe-capability control. The carrier-versus-causal dissociation, the family-dependent sign, and the positive-control argument are, to our knowledge, new.

Our work also connects to studies of uncertainty in language models. Semantic entropy estimates predictive uncertainty by clustering semantically equivalent generations [13, 7], and this signal is recoverable from a single generation’s hidden states by a trained probe [12]. These works target semantic uncertainty over generations and select probes by supervision, whereas we study token-level output entropy and a fixed canonical neuron set; we additionally show that correlation-selected and null-space-selected neurons form disjoint sets (Section 4.5)—two operational definitions rather than a correct and a mistaken one. That uncertainty is linearly decodable from hidden states [12] is consistent with our $R^2_{\text{full}} \approx 0.7$; the contribution here is to show that this decodable direction is dissociated from the canonical neurons.

Two recent studies bear directly on the results of Section 4.7. [14] recover a distributed correctness signal from residual activations via a regularized probe rather than individual units, arguing that

single sparse features have negligible causal effect and that the relevant information is a collective property of the activation space; they steer with the learned direction and evaluate its effect through downstream accuracy and calibration gains. As reported, their evaluation does not include the anti-triviality controls used here (a matched random subspace, a matched PCA subspace, and a matched global temperature), and their target quantity is response correctness rather than output entropy. We intervene on the residual subspace directly via interchange and causal necessity against those three matched controls, and locate a low-rank, not merely distributed, carrier in one architecture. [8] independently report, in the unrelated context of on-policy self-recognition, that a decodable direction tracking predicted-token entropy has near-zero causal effect under steering (slope 0.01–0.04 against a causal reference of 1) while a separate decodable surprise direction is fully causal; this is the same qualitative dissociation we report in Section 4.7, obtained by a different method (centroid steering versus interchange-transfer) on a single frontier model rather than across architectures, and we read the convergence as evidence that prediction-versus-intervention dissociations of this kind are not an artifact of either method. A third, differently structured dissociation is reported by [15]: verbalized confidence and a separate calibration signal are each linearly decodable from a model’s activations, but the two directions are geometrically orthogonal to one another, across three open-weight models and four datasets. That finding dissociates two decodable quantities from each other; ours dissociates a single quantity’s decodability from its causal use (immediately above) and, separately, a mechanism’s causal influence from whether it linearly carries the variance it influences (Section 4.2)—a different pair of axes, applied to output entropy rather than confidence, with no direct overlap in claims. Separately, [20] study entropy neurons’ role in suppressing context-copying under parametric-versus-contextual knowledge conflicts, using the same selection criteria as [19], and [5] find that canonical entropy neurons have similar activation distributions between base and post-trained (SFT/RLHF) checkpoints of the same model and, specifically, do not account for the confidence differences observed between them—evidence that whatever mechanism sets confidence post-training, it is not routed through these neurons. Both studies are complementary to, rather than in conflict with, the carrying-versus-influence dissociation and the continued-pretraining evidence of Section 4.9: post-training and continued pretraining are different interventions, and neither study measures whether the neurons *carry* entropy’s variance in the sense of Section 3.4.

3 Method

3.1 Notation and preliminaries

Let a model with L layers process a token sequence. Write $r_t \in \mathbb{R}^{d_{\text{model}}}$ for the residual stream at the final layer $L_f = L - 1$ and position t , $a_t \in \mathbb{R}^{d_{\text{mlp}}}$ for the final-MLP post-activation, $W_{\text{out}} \in \mathbb{R}^{d_{\text{mlp}} \times d_{\text{model}}}$ for the MLP output weights, and $W_U \in \mathbb{R}^{d_{\text{model}} \times |V|}$ for the unembedding. Output entropy at position t is

$$H(x_t) = - \sum_{v \in V} p_t(v) \log p_t(v), \quad p_t = \text{softmax}(z_t), \quad (1)$$

with z_t the logits. We use three terms precisely. *Influence* is the causal change in H when entropy neurons are ablated (Section 3.5). *Carrying*, used interchangeably with *encoding*, is the share of the variance of H that a neuron set linearly explains, i.e. the held-out R_{EN}^2 of Section 3.4. *Decodability* is the held-out R_{full}^2 of H from the full residual-stream input. Decodability concerns the residual stream; carrying concerns a specific neuron set; the two are not the same.

3.2 Identifying entropy neurons

For neuron j we compute a null-space proxy: the logit effect of its output direction per unit of residual norm, in the spirit of logit-lens projection [16, 1],

$$\nu_j = \frac{1}{1 + \frac{\|\widetilde{W_{\text{out}}[j]} W_U\|_2}{\|\widetilde{W_{\text{out}}[j]}\|_2}}, \quad (2)$$

where $\widetilde{(\cdot)}$ denotes mean-centering over the vocabulary. This is a monotone proxy for the fraction of the output weight lying in the unembedding null space; we regard it as a proxy, not an exact measure. Four absolute gates then apply: ν_j above the 98th percentile over all neurons; clip-ablation changing entropy at high-activation positions by more than 0.02 nats; the argmax token flipping on fewer than 1% of positions; and the ablation moving the residual norm by at least 0.30 (an absolute change, unnormalized across models). These reproduce the canonical neurons of prior work (GPT-2 small, layer 11 neuron 2378; Pythia-410M, layer 23).

We disclose a circularity and bound its reach. Because one gate selects for a causal entropy effect, “the neurons are causal” is partly built into selection; we therefore treat only the *magnitude*, *sign*, and *contrast against a matched control* of the causal effect as informative (Section 4.2). This circularity does *not* reach the carrier result: the carrier probe (Section 3.4) measures how much entropy variance the selected neurons explain and never uses the causal gate as a feature, so a model can pass selection yet carry nothing. The residual-norm gate is unnormalized across models, which affects how many neurons are admitted but not the probe-based carrier verdict.

Finally, selecting neurons by their correlation with entropy yields a different top set than Eq. 2 in nearly every model (Section 4.5).

3.3 A sign-invariant attribution measure

To describe how distributed the upstream drive of the entropy neurons is, let $c_i = |(W_{\text{in}})_i a_i|$ be the absolute contribution of upstream unit i to the entropy-neuron pre-activation, with a_i its activation and $(W_{\text{in}})_i$ its weight into the neuron, and let $c_{(1)} \geq c_{(2)} \geq \dots$ denote the sorted values. The cumulative profile and its p -coverage point are

$$F(k) = \frac{\sum_{i \leq k} c_{(i)}}{\sum_i c_i}, \quad k_p = \min\{k : F(k) \geq p\}. \quad (3)$$

On synthetic data with a known number of planted contributors (Appendix A, Figure 7), recall of the planted set is 100% at the true sparsity, $k_{0.5}$ grows monotonically with planted sparsity, and the profile is identical for positive, negative, and mixed signs, confirming sign-invariance; under a no-structure null, $k_{0.5}$ saturates at the maximum. This validates recall and sign-invariance, not precision; we do not claim a measured false-positive rate on real networks.

3.4 Probing whether the neurons carry entropy variation

We predict per-token H from the final-MLP input by ridge regression on document-disjoint train/test splits. Features are standardized column-wise, $\tilde{X} = (X - \mu) / \sigma$ using training-fold statistics; with target y ,

$$\hat{\beta} = \arg \min_{\beta} \|y - \tilde{X}\beta\|_2^2 + \lambda \|\beta\|_2^2, \quad R^2 = 1 - \frac{\|y_{\text{te}} - \tilde{X}_{\text{te}}\hat{\beta}\|_2^2}{\|y_{\text{te}} - \bar{y}_{\text{te}}\|_2^2}, \quad (4)$$

with $\lambda = 1$ and R^2 evaluated out of sample. Point estimates use the disjoint split; intervals use a 1000-replicate bootstrap that resamples within each side, so no document appears in both train and test within a replicate (replicates with fewer than three unique test documents are skipped; this did not occur). We compare three feature sets on identical splits: the entropy neurons (R_{EN}^2), a same-sized disjoint random-neuron set (R_{rand}^2), and—as positive controls—the top correlation-with-entropy neurons and the top principal components of the input. The correlation-with-entropy control selects on the training fold only; an earlier version selected on all evaluation documents, leaking test labels into selection, and is corrected here.

Corrected gap statistic. An earlier version reported the carrier margin as $R_{\text{EN,lo}}^2 - R_{\text{rand,hi}}^2$, a difference of independent marginal tails; because the two are computed on the same documents and positively correlated, this overstates the width and yields spuriously negative margins. We instead form, within each of $B = 1000$ bootstrap replicates indexed by b on the *same* resampled documents, the paired difference g_b , and summarize it by the α -percentile Q_α and the one-sided mass below zero:

$$g_b = R_{\text{EN}}^2(b) - R_{\text{rand}}^2(b), \quad \text{gap}_{\text{lo}}, \text{gap}_{\text{hi}} = Q_{2.5}(\{g_b\}), Q_{97.5}(\{g_b\}), \quad p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[g_b \leq 0]. \quad (5)$$

Verdict. With seeds $s \in \{0, 1, 2\}$ and $\text{gap}_{\text{lo}}^{(s)}$ the lower bound on seed s ,

$$\text{model is } \begin{cases} \text{carrier} & \text{if } \min_s \text{gap}_{\text{lo}}^{(s)} > 0, \\ \text{near-miss} & \text{if } \exists s : \text{gap}_{\text{lo}}^{(s)} > 0, \\ \text{non-carrier} & \text{otherwise.} \end{cases} \quad (6)$$

The worst-seed requirement is the rule that retires a self-corrected artifact (Section 4.6).

3.5 Causal protocol

We clip- and mean-ablate the entropy neurons and measure the change in H at high-activation positions, with document-level intervals, against a disjoint random-neuron baseline. We report clip as the defensible estimate and mean (inflated by the LayerNorm feedback loop) only relatively. Causal intervals are over at most 12 documents (the injection test of Appendix C uses 8); the wide Pythia and OPT intervals are a direct consequence of this small n . We flag one model: for OPT the random baseline is not near zero (≈ 0.32 nats), so OPT’s large effect has low signal-to-noise and we do not use it as a clean example.

4 Results

4.1 Entropy is decodable everywhere

The gates recover entropy neurons in 11 of the 13 models; Pythia-160M and TinyStories-33M, both fully trained, yield none. In every model that contains them, the input to the final MLP predicts per-token entropy with a held-out R_{full}^2 between 0.55 and 0.74 (Figure 1, Table 1), including the models whose entropy neurons prove not to carry it. Entropy is thus linearly recoverable from the residual stream throughout the model set. Whether it is recoverable from the canonical neurons in particular is a separate question.

Model	n_{EN}	R_{full}^2	R_{EN}^2 [95% CI]	R_{rand}^2	gap_{lo}	p	corrH	verdict
GPT-2	6	0.70	0.375 [0.33, 0.43]	-0.035	0.197	<.001	0.47	carrier
GPT-2-medium	6	0.74	0.395 [0.35, 0.45]	0.045	0.191	<.001	0.47	carrier
GPT-2-large	5	0.60	0.206 [0.15, 0.25]	0.027	0.116	<.001	0.18	carrier
DistilGPT-2	6	0.65	0.401 [0.33, 0.46]	0.024	0.279	<.001	0.26	carrier
Pythia-410M	6	0.65	0.022 [-0.03, 0.06]	0.013	-0.132	.579	0.42	no
Pythia-1B	6	0.55	0.024 [-0.00, 0.04]	0.032	-0.066	.374	0.30	no
Pythia-1.4B	6	0.62	0.105 [0.04, 0.15]	0.038	-0.037	.073	0.49	no*
GPT-Neo-125M	4	0.63	-0.000 [-0.02, 0.02]	-0.008	-0.157	.750	0.39	no
BLOOM-560M	1	0.63	0.012 [0.00, 0.02]	0.051	-0.046	.211	0.16	no
Qwen2.5-0.5B	5	0.64	0.194 [0.02, 0.32]	0.015	0.010	.019	0.31	no*
OPT-125M	4	0.62	0.013 [-0.02, 0.03]	0.077	-0.100	.471	0.34	no

Table 1: Dissociation across the 11 models with entropy neurons (Pythia-160M and TinyStories-33M, both fully trained, have none and are omitted). R_{full}^2 : held-out decodability of entropy from the final-MLP input. $R_{\text{EN}}^2/R_{\text{rand}}^2$: held-out variance carried by the entropy neurons vs. a same-sized random set, with the 95% bootstrap CI on R_{EN}^2 . gap_{lo} and p : lower bound and one-sided mass of the paired-bootstrap gap (Eq. 5). corrH: correlation-with-entropy positive control (train-fold selection). Verdict per Eq. 6; * denotes a near-miss (the paired criterion passes on at least one seed but not the worst seed under the worst-seed rule). All quantities are held-out; CIs are 2.5/97.5 bootstrap percentiles. R_{full}^2 is a point estimate (its bootstrap behaviour is discussed in Section 5). Activation- and entropy-variance descriptors are in Table 4.

4.2 Causal but not carrier

Ablation of the entropy neurons alters output entropy in every model that contains them. The clip-ablation shift has a confidence interval excluding zero in all eleven: 0.04 to 0.10 nats in the GPT-2 lineage, 0.49 to 2.52 nats in Pythia, GPT-Neo, and OPT, and an intermediate 0.04 to 0.25 nats in BLOOM and Qwen, against a random-neuron baseline near zero in every model but OPT, whose non-zero baseline (Section 3.5) leads us to set it aside. Because one selection gate rewards a causal effect (Section 3.2), the existence of this effect is in part a consequence of selection; its magnitude and sign are not, and the analysis rests on those.

This causal influence does not, however, extend to carrying. Outside the GPT-2 lineage the neurons do not account for the variation of entropy that the model exhibits. The entropy-neuron probe attains an R_{EN}^2 of 0.375, 0.395, 0.206, and 0.401 across the four GPT-2 models, and in each its paired-bootstrap margin over the random baseline is bounded away from zero (lower bounds 0.197, 0.191, 0.116, and 0.279; the paired difference is positive in all one thousand replicates) and stable across the three seeds. In no other family is the margin positive. The positive correlation between R_{EN}^2 and R_{rand}^2 across replicates, between 0.06 and 0.13, motivates the paired comparison: the marginal-difference statistic used previously treats these correlated quantities as independent and understates the margin.

This localization and the dissociation that underlies it differ in evidential strength. The dissociation itself—a large causal effect together with a carried variance indistinguishable from the random baseline—is present in all seven non-GPT-2 models with entropy neurons and does not depend on any one of them. The narrower claim, that carrying is confined to the GPT-2 family, rests on four checkpoints sharing an architecture and a pre-training corpus, one a distillation of another, and we regard it as more tentative.

A third dissociation concerns the direction of the causal effect, which varies by family: ablation

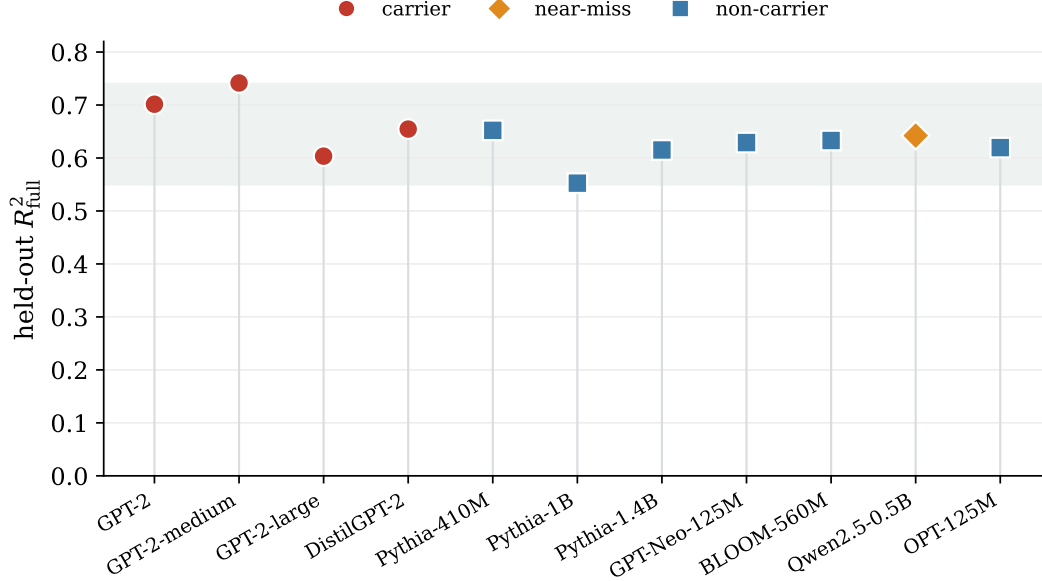


Figure 1: Output entropy is linearly decodable from the final-MLP input in every model that has entropy neurons (EN). Each point is the held-out R^2_{full} for one model, marker shape and colour giving its verdict; the shaded band marks the observed range, 0.55 to 0.74. Decodability is uniformly high and does not depend on the carrier verdict.

raises entropy in Pythia, GPT-Neo, and OPT and lowers it in GPT-2, BLOOM, and Qwen (Figure 8). The sign does not track the carrier split—BLOOM and Qwen lower entropy without carrying its variation—so we report it descriptively rather than as a second functional dimension; it may reflect a sign convention in the weights together with the clip operation.

The families with the largest intervention effect thus carry the least variance, and the family that carries variance exhibits the smallest effect. We refrain from describing this as an anti-correlation. It is a contrast between two groups rather than a continuous trend: the carrying group is effectively a single lineage, Qwen departs from the pattern within the other group, and the causal magnitudes are not comparable across models.

The variance is instead recoverable from the direction of the residual stream more readily than from its norm. On the four anchor models, a probe on the normalized residual attains $R^2 \approx 0.7$, whereas one restricted to the LayerNorm scale remains at or below 0.17 (Figure 3). The direction probe has d_{model} features and the norm probe one, so the appropriate conclusion is not that a single direction carries entropy but that the signal is distributed: a PCA probe reaches approximately 0.50 with four components and increases only gradually thereafter, plateauing near 256. A signal distributed across a direction the neurons do not span, transmitted to the output through the norm they set, accounts for the combination of a strong causal effect and little carried variance.

4.3 The probe is capable

A low R^2_{EN} is consistent with two interpretations: that the neurons carry little entropy variance, or that no small neuron set permits its linear recovery in these models. The positive controls distinguish them. In Pythia and GPT-Neo, a probe constructed from the neurons most correlated with entropy attains R^2 between 0.31 and 0.49, and a PCA probe reaches 0.38 in Pythia-410M,

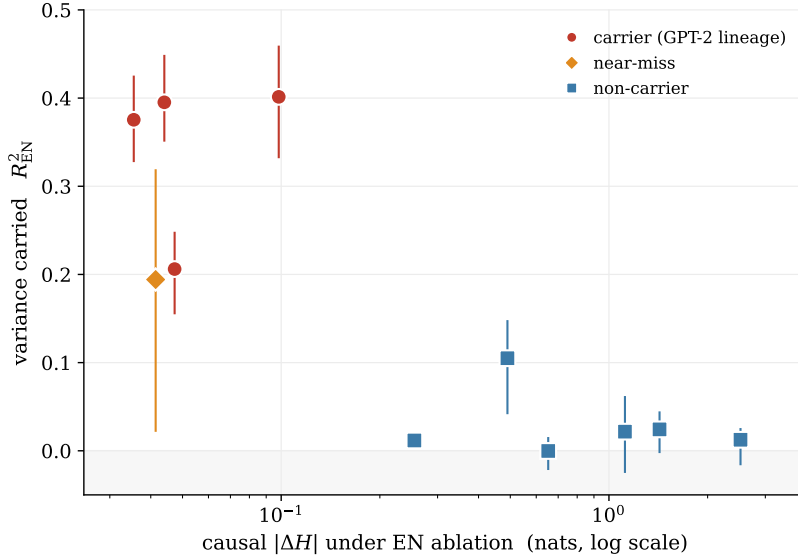


Figure 2: A strong causal effect coincides with little carried variance. Carried variance R^2_{EN} (y-axis, over 60 evaluation documents) against causal $|\Delta H|$ under ablation (x-axis, log scale, over ≤ 12 documents). Carriers (GPT-2 lineage) lie upper-left, non-carriers lower-right, near-misses in amber; verdict is encoded by colour and marker shape (legible in grayscale). The shaded band below zero marks negative R^2_{EN} (worse than predicting the mean). The wide horizontal spread of the non-carriers reflects the small causal-document count. Intervals are 2.5/97.5 bootstrap percentiles. EN: entropy neurons.

whose entropy-neuron probe is itself at the random baseline (Figure 4); the highest PCA value, 0.55, belongs to GPT-2-medium, a carrier, and does not bear on this comparison. Entropy is linearly accessible in these models but not through the entropy neurons. Two models qualify this control. BLOOM contains a single entropy neuron, and both of its controls are degenerate, at 0.16 and 0.012, so probe-capability cannot be established and we rely on its full-residual decode of 0.63. OPT’s PCA control is likewise degenerate, at 0.026, while its correlation control, at 0.34, supports the conclusion.

4.4 A single lineage, not the tokenizer

The carriers are precisely the GPT-2 family, including the distilled model. The most immediate alternative explanation, the tokenizer, is not supported: GPT-Neo-125M uses the GPT-2 tokenizer and its vocabulary of 50,257 tokens yet carries nothing. This inference rests on the vocabulary size and the established identity of the tokenizer; the per-model hash we compute does not reliably certify tokenizer identity, for reasons discussed in Section 5. What remains confounded with the lineage is the GPT-2 architecture, the WebText pre-training corpus, and the out-of-distribution status of our evaluation text for GPT-2 relative to its in-distribution status for Pythia and GPT-Neo. On pretrained checkpoints these three factors covary and cannot be separated.

4.5 Two criteria, different neurons

The two criteria for identifying an entropy neuron select different neurons. The top-five sets ranked by correlation with entropy and by the null-space criterion share no members in nine of the eleven

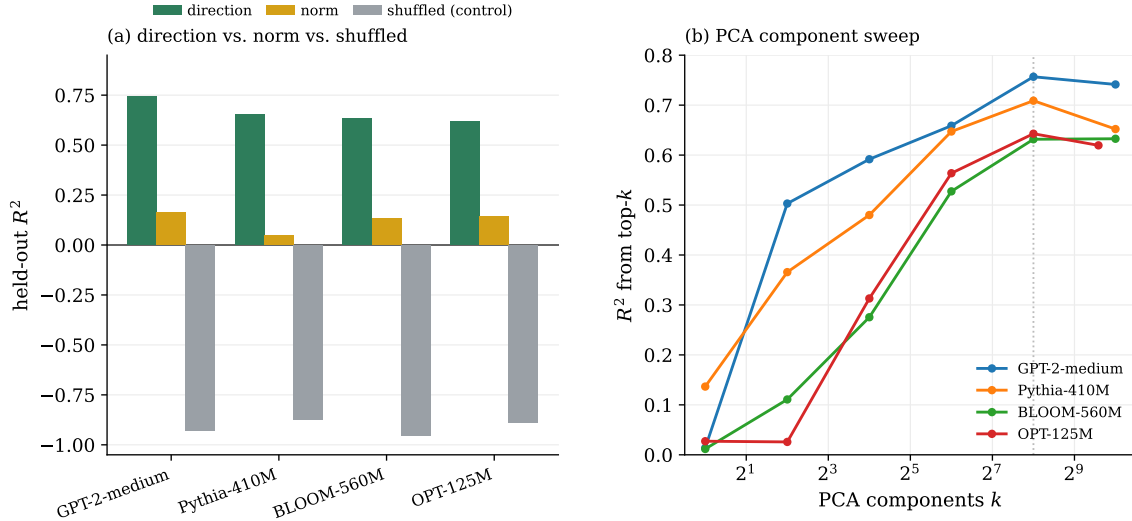


Figure 3: Mechanism, on the four anchor models. (a) Held-out R^2 for entropy from the residual *direction*, from the LayerNorm *norm*, and from a *shuffled* control (≈ -0.9 , confirming the direction signal is real). (b) R^2 from the top- k principal components (PCA): the signal saturates only near $k = 256$ (dotted), so it is distributed rather than low-dimensional.

models and one member in the remaining two (Table 6). They constitute distinct operational definitions addressing distinct questions, which should be kept in mind when interpreting a result obtained under one criterion as a statement about the other.

4.6 Robustness and negative results

Two non-GPT-2 models warrant closer examination, because their point margin is positive despite a non-carrier verdict. Qwen2.5-0.5B has an R_{EN}^2 of 0.194 and a point margin of 0.179; it satisfies the paired criterion on two of three seeds, with the paired difference non-positive in 1.9%, 3.5%, and 28.7% of replicates, but not on the third, where R_{EN}^2 falls to 0.067. Pythia-1.4B is lower, at R_{EN}^2 0.105 and margin 0.067, satisfying the criterion on one seed only (paired difference non-positive in 7.3%, 10.7%, and 0.4%). The worst-seed rule (Eq. 6) classifies both as non-carriers; we report them as near-misses rather than as unambiguous negatives. Figure 9 shows this seed-to-seed spread against the tight clustering of the four carriers.

Correction for multiple comparisons reinforces this classification. Across eleven models, a Bonferroni threshold is approximately 0.0045; the four carriers, at $p < 0.001$, satisfy it, whereas the near-misses do not, including Qwen at its best-seed p of 0.019. Evaluated on its strongest seed alone, without the worst-seed requirement, Qwen would satisfy the paired criterion; the worst-seed rule and the multiplicity correction exclude it independently. This is the least robust component of the single-lineage claim, and we report it as such.

The worst-seed rule originates in a corrected error. An earlier analysis estimated Pythia-1.4B’s R_{EN}^2 near 0.40 and interpreted it as carrying that emerges with scale. The estimate did not survive disjoint selection and evaluation pools with leak-free bootstrapping: it fell to 0.105, with per-seed values of 0.105, 0.094, and 0.220 and a margin that did not hold. We attribute the original value to selection and leakage, and the requirement that every seed clear the threshold follows from this correction.

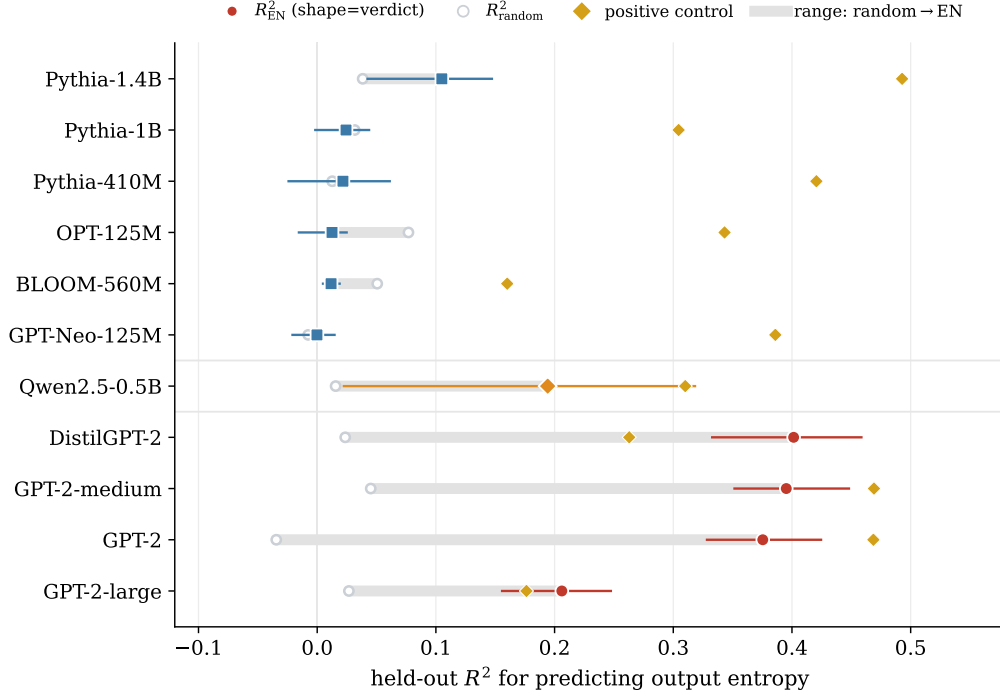


Figure 4: Entropy is decodable, but not from the entropy neurons. Per model, the entropy-neuron probe R^2_{EN} (filled, shape = verdict, with CI), the random-neuron baseline R^2_{random} (open), and the positive control (diamond; corrH, the correlation-with-entropy probe). Models grouped by verdict; within each group the grey bar spans the random-to-EN range. The positive control is high in the non-carriers, where R^2_{EN} is at baseline.

The near-zero R^2_{EN} outside the GPT-2 lineage might be attributed to near-constant neurons, on which any probe would be uninformative. The data are inconsistent with this account. Carrying is, if anything, inversely related to activation variance: the two models with the most variable entropy neurons, Qwen and Pythia-410M at 50.6 and 15.9, are a fragile near-miss and a non-carrier respectively, whereas DistilGPT-2, whose entropy neurons are least variable at 0.46, is among the strongest carriers. The probe additionally standardizes its inputs, so low variance alone cannot reduce R^2 to zero.

Carried variance is likewise not a function of the range of entropy itself. The per-token entropy variance of the carriers, 3.43 to 4.09, overlaps the non-carrier range of 2.90 to 3.79, and GPT-Neo, a non-carrier, is among the highest at 3.74. The broader out-of-distribution confound remains open (Section 5), but this form of it is not supported by the data.

We were unable to determine why the GPT-2 lineage carries entropy; no candidate we examined separates the two groups (Figure 5). The fraction of the ablation effect carried by the tangential, norm-changing channel overlaps between groups—GPT-2-medium and Pythia-410M coincide at 0.80—so it does not mark the carriers. A geometry probe of null-space alignment is partly circular, since the neurons are selected for writing to the null space, and the appropriate control—neurons passing the same gate but not selected—diminishes the apparent signal; the alignment is in any case no higher for carriers, its largest value belonging to GPT-Neo, a non-carrier. A causal-direction test, which injects the neurons’ null-space write direction and compares its effect on entropy with that of random null directions, is large for only two of the four carriers (GPT-2-large and DistilGPT-2,

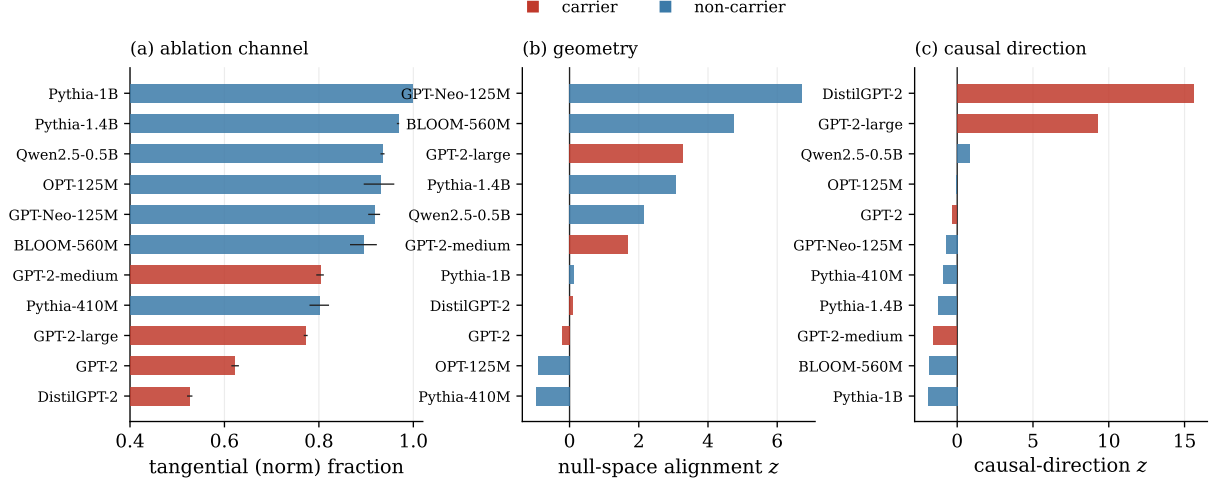


Figure 5: No candidate explanation separates carriers (red) from non-carriers (blue). (a) The tangential (norm-changing) fraction of the ablation effect overlaps between groups (GPT-2-medium and Pythia-410M coincide at 0.80). (b) Null-space geometric alignment (z -score) is not higher for carriers; the largest value is GPT-Neo, a non-carrier. (c) The causal-direction z -score is large for only two of the four carriers. Panels have independent x-scales; the comparison of interest is carrier vs. non-carrier within each panel, not bar lengths across panels.

at 9.3 and 15.6), negative for the other two (GPT-2 and GPT-2-medium, at -0.31 and -1.58), and near zero or negative across the non-carriers, with a small positive value for Qwen the only exception. We draw no general mechanism from these, consistent with the path from the null space through the norm to the entropy being generic rather than specific to the carriers.

4.7 A distributed causal carrier outside the entropy neurons

Section 6 of an earlier version of this work posed identifying the distributed direction that does carry entropy—and testing whether steering it controls output entropy where the canonical neurons do not—as the natural next step. We take up that question directly.

Method. For a held-out subspace $U \in \mathbb{R}^{d_{\text{model}} \times k}$ at the final residual site r_t (input to ln_final), we define the *interchange* operator $A_{\text{patch}} = A_b + ((A_s - A_b)U)U^\top$, which replaces the base sequence’s U -projection with the source sequence’s, and measure the fraction of the mean entropy gap it closes, $\text{tr} = (H(A_{\text{patch}}) - H(A_b)) / (H(A_s) - H(A_b))$ (Table 2’s tr_{rand} , tr_{PCA} : tr against each matched control). We additionally measure *causal necessity* nec (Table 2’s nec_{PCA}): the drop in the model’s real output-entropy variance when U ’s projection is pinned to its corpus mean, $A - (AU - \overline{AU})U^\top$. U is learned on a disjoint training split as the ridge-supervised direction for H (column one) together with $k - 1$ further directions spanning the $|dH|^2$ -weighted activation scatter, orthogonalized; k is selected on a held-out validation split by the value that maximizes the validation ridge probe R^2 of H from U —a prediction-side criterion, decoupled from the gated PCA/random/temperature contrasts, which are computed only at the selected k and never used to choose it. We report transfer and necessity against three matched controls on identical splits—a rank-matched random subspace, the top- k principal components (the anti-triviality control: U must beat unsupervised PCA, or the result reduces to “entropy is distributed in the residual stream,” already established in Section 4.1),

Model	$\text{tr}_{\text{rand,lo}}$	$\text{tr}_{\text{PCA,lo}}$	$\text{nec}_{\text{PCA,lo}}$	EN ovl.	MDE	verdict
GPT-2 (control)	0.365	−0.084	0.016	0.91	0.162	EN-carrier
GPT-2-medium	0.200	−0.069	0.013	0.65	0.183	EN-carrier
DistilGPT-2	0.355	−0.031	0.062	0.90	0.129	EN-carrier
GPT-Neo-125M	0.162	0.132	0.379	0.008	0.092	PCA-carrier
GPT-Neo-1.3B	−0.040	−0.158	0.004	0.52	0.242 [‡]	non-carrier
Pythia-410M	−0.146	−0.135	−0.028	0.11	0.257 [‡]	non-carrier
Pythia-160M	0.387	−0.269	−0.104	–	0.161 [‡]	non-carrier
Pythia-70M	0.336	−0.140	−0.054	–	0.138	non-carrier
OPT-125M	0.001	−0.104	0.141 [†]	0.09	0.019	non-carrier
BLOOM-560M	0.222	−0.362	0.067	0.36	0.154 [‡]	non-carrier

Table 2: Interchange-transfer and causal-necessity margins at the final residual site, worst-seed family-wise lower bounds, across four acceptance-verified, prereg-checked runs (torch 2.7.1, `data_hash da5f6242b06fa229`). U is the learned interchange subspace defined in the Method paragraph above. tr_{rand} and tr_{PCA} : raw-transfer margin over a rank-matched random subspace and over top- k PCA (positive and > 0.15 is the per-model transfer criterion). nec_{PCA} : causal-necessity margin over PCA- k ablation, the anti-triviality control on the causal side. EN ovl.: overlap of U with the canonical entropy-neuron write-subspace (“–” where a model has no entropy neurons under Section 3.2’s criterion, making the overlap undefined). MDE: the paired-bootstrap minimum detectable effect, same units as tr_{rand} . [†]OPT’s positive necessity margin is a control artifact—PCA- k ablation is itself anti-helpful there (−0.13), while OPT’s own absolute necessity is ≈ 0.01 ; we do not read it as evidence of a carrier. [‡]These four non-carrier verdicts have $\text{MDE} > 0.15$: the study is underpowered to rule out a transfer margin at our criterion, so “non-carrier” here is an absence of detected evidence, not evidence of absence. The other two non-carriers (Pythia-70M, OPT-125M) are well-powered nulls, $\text{MDE} < 0.15$; OPT’s $\text{MDE} = 0.019$ is the tightest in the panel. *EN-carrier*: carries via the canonical entropy neurons (Table 1), not PCA-distinct. *PCA-carrier*: distinct from PCA at every tested k , entropy-neuron-orthogonal. Exactly one of nine non-anchor models is a PCA-carrier.

and a global scalar temperature matched to the same mean shift—together with the canonical entropy-neuron write-subspace as a fourth, descriptive contrast. Bounds are worst-seed, family-wise (Bonferroni) paired-bootstrap lower bounds over three seeds, at the smallest-effect-size-of-interest ($\text{SESOI} = 0.15$) raw-transfer margin used throughout this paper. Full protocol, synthetic self-check, and prereg verification (bytecode method-hash, frozen before each run) are released with the artifacts.

An EN-orthogonal carrier in GPT-Neo-125M. Across three seeds, GPT-Neo-125M has a one-dimensional ($k = 1$), causally sufficient, bidirectional entropy carrier that is almost entirely disjoint from the canonical entropy neurons (write-subspace overlap 0.008, against 0.65–0.91 in the GPT-2 lineage). It beats the random-subspace control (margin lower bound 0.162), beats PCA- k at every tested $k \in \{1, \dots, 256\}$ (Figure 6), and its causal-necessity margin over PCA ablation is the largest in the panel (0.379, $p \approx 0$). Raising and lowering entropy are both available from the same direction with comparable magnitude (asymmetry 0.13–0.16 across seeds), unlike every other model in the panel, where interchange behaves as a one-directional subspace swap (asymmetry ≈ 1.0): patching the source projection in reproduces a large mean shift, but patching it out does not reverse it by a comparable amount. We are not aware of a prior report of a low-rank, entropy-neuron-orthogonal, bidirectionally controllable causal entropy carrier outside the GPT-2 lineage.

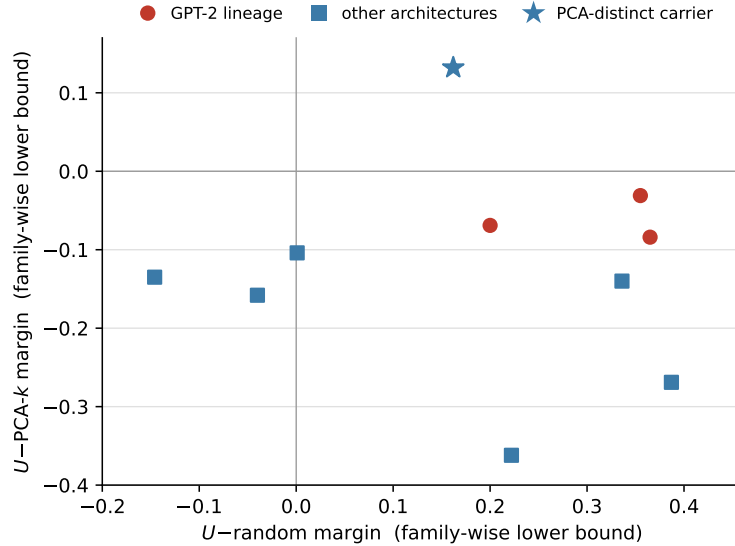


Figure 6: Beating a random subspace is common; beating PCA- k is not. Raw-transfer margin over random (x-axis) versus over PCA- k (y-axis), worst-seed family-wise lower bounds. GPT-2-lineage models (circles) carry via the entropy neurons but sit below the $y = 0$ line: an unsupervised principal subspace of the same rank explains their transfer margin at least as well as the learned subspace U (defined in the Method paragraph above) does. GPT-Neo-125M (star) is the only model above it.

Prediction is not intervention. In every model in the panel, a ridge probe on the same final-residual site recovers entropy at $R^2 \approx 0.51\text{--}0.75$ already at $k = 1$ (lowest in Pythia-70M, highest in GPT-2-medium) and is flat in k thereafter—the predictive direction is one-dimensional and equally readable everywhere. Yet low- k *causal* controllability—beating PCA- k under interchange—exists only in GPT-Neo-125M. A direction can fully predict output entropy while being causally near-inert: in GPT-Neo-1.3B, for example, the $k = 1$ raw transfer is negative. This dissociation between decodability and controllability at the level of a single residual direction is independent of, and complements, the carrying-versus-influence dissociation of Section 4.2; a closely related observation, that a decoded quantity need not be the causal lever for it, was reported concurrently and independently in a different context (self-recognition in post-trained models) by [8], who found a decodable entropy-related direction with steering slope 0.01–0.04 against a causal reference of 1, while a separate decodable quantity (input surprise) was fully causal. We extend the same dissociation to the entropy-neuron site specifically and show it holds across architectures, not only within one frontier model.¹

The carrier does not survive scaling within its own lineage. A natural hypothesis is that GPT-Neo-125M’s carrier reflects its architecture (learned absolute positional embeddings, tied embeddings, alternating local/global attention) or its pretraining corpus (The Pile). Both are held fixed in GPT-Neo-1.3B, which reproduces neither: its transfer margin over PCA is negative at its family-wise lower bound (−0.158) and at seven of the nine tested k , the two exceptions ($k = 32$,

¹An independently coded, PCA-based pipeline qualitatively reproduces both patterns on an overlapping model set (a held-out probe flat in k from $k = 1$; a rank-one subspace that is both decodability-sufficient and causally necessary only in GPT-Neo-125M), using a different pool size and without a pre-registered blind holdout. We treat this as a robustness cross-check, not independent confirmation, and its numbers do not appear in our tables.

$k = 128$) being point estimates no larger than 0.06; its necessity margin over PCA is a statistical tie (0.004), its interchange behaves as a one-directional subspace swap, and the partial correlation between the carrier-direction’s projection energy and output entropy (controlling for position and log-frequency) *changes sign* between the two checkpoints (-0.45 at 125M versus $+0.47$ at 1.3B). This is not a weaker version of the same object; it is a qualitatively different one. Because architecture and training corpus are identical across the pair, this rules out both as the sole explanation and narrows the candidate mechanism toward something that covaries with scale or capacity within a fixed recipe (Section 4.8).

Extended panel and ceiling. Across four independent, prereg-verified, environment-checked runs covering nine non-anchor models (Table 2, including the two GPT-2-family relatives tested under this protocol), exactly one model shows the full clean pattern; the decisive test (transfer over PCA- k across the entire k -grid, Figure 6) was run in every case, so the absence elsewhere is measured rather than an artifact of an undersized k -grid or an unlucky seed. Power is not uniform: four of the six non-carriers are underpowered at our transfer criterion (Table 2), so their absence should be read as undetected rather than ruled out; the two well-powered nulls (Pythia-70M, OPT-125M) are the only ones we treat as confident evidence of absence. We read the graduated verdict as WEAK under the same worst-seed, family-wise discipline as Table 1: a real, replicated, controlled-for-triviality positive result in one architecture, not (yet) a second axis of universal structure. We report this as the measured ceiling of the present panel rather than extend it further, since additional undirected model additions after four rounds of testing would constitute a forking-path search rather than a confirmatory one.

4.8 Architectural axes across the combined panel

The GPT-Neo-125M/1.3B pair of Section 4.7 is a sharper instrument than the within-lineage battery of Section 4.6 for the same question, because architecture and pretraining corpus are held *exactly* fixed across the pair rather than merely shared within a family. We extend the architectural comparison across the combined panel (Table 1 $\cup$ Table 2: fifteen distinct models, thirteen of which appear in Table 1, with GPT-Neo-1.3B and Pythia-70M tested only under the protocol of Table 2) on five documented axes—positional-encoding scheme, embedding tying, attention pattern, normalization placement, and pretraining corpus—verified against each architecture’s released configuration and technical report rather than measured with the in-house tying detector found to be faulty below.

No single axis separates carriers from non-carriers cleanly in both directions under either criterion. Under the EN-carrying criterion of Table 1, where all thirteen of its models are measured, two axes give a mostly one-directional signal with one exception each: every model with a non-learned positional scheme (rotary in the three Pythia checkpoints and in Qwen2.5-0.5B; ALiBi in BLOOM) is a non-carrier, except that Qwen is the near-miss of Section 4.6 (a positive point margin failing the worst-seed criterion), not a clean negative; and every model with untied embeddings (the Pythia checkpoints, via their GPT-NeoX-style architecture) is likewise a non-carrier, except that Pythia-1.4B is the other near-miss of that section. Under the distinct-direction criterion of Section 4.7—tested only on Table 2’s nine-model panel, which does not include Qwen or the larger Pythia checkpoints—the same two axes give a clean 0-of-4 and 0-of-3 signal. Learned positional embeddings and tied embeddings, however, are present in both carriers and non-carriers under either criterion (including GPT-Neo-1.3B) and so are not themselves sufficient. Local/global alternating attention is shared by GPT-Neo-125M and GPT-Neo-1.3B and therefore cannot, by construction, distinguish the pair. Pretraining corpus is the axis most directly falsified by the pair: GPT-Neo-125M, trained on

The Pile, is the panel’s only distinct-direction carrier, while GPT-Neo-1.3B and the three Pythia checkpoints—also Pile-trained—are not, which is inconsistent with a simple corpus-identity account of carrying and complements the per-domain evidence of Section 4.9.

We read this as ruling out each single-axis hypothesis we could test on existing checkpoints, not as evidence that architecture is irrelevant, and we note the two near-misses above as a reason to treat the positional-encoding and embedding-tying patterns as suggestive rather than exceptionless. A two-way, matched-compute pretrain that varies scale within a fixed architecture and corpus, rather than varying architecture or corpus at fixed scale, is the comparison our own evidence now points to, and we did not run it.

4.9 Per-domain evidence on the domain confound

Section 5 identifies the evaluation-domain match between the GPT-2 lineage (WebText) and our Pile-based evaluation corpus as the most consequential open confound, and proposes a per-domain control as the most readily available unperformed measurement. We ran that control via continued pretraining (CPT): starting from GPT-2 and from GPT-Neo-125M—both 125M-scale and sharing the GPT-2 tokenizer, so tokenizer and size are held constant by design—we continue training each on its native domain (a same-domain control arm) and on the other’s domain (a swapped arm), and compare the resulting shift in entropy-neuron carrying via a difference-in-differences (DiD) on the paired, frozen-EN gap statistic at each model’s native evaluation domain (the pre-registered primary endpoint), against a random-neuron placebo and an EN-survival gate. Both arms of both models were run to a token budget of 5×10^7 at a single seed; the protocol’s own pre-registration explicitly flags ≥ 3 seeds per arm as required for a main-track-grade confirmatory run, which we have not yet performed, so we report point estimates from a single-seed pilot rather than a confirmed result.

The domain-transfer effect does not survive as a clean signal. For GPT-2 (native domain WebText/OWT, swapped to Pile), the frozen-EN DiD at the primary endpoint is -0.034 ; for GPT-Neo-125M (native Pile, swapped to WebText/OWT), it is -0.001 , indistinguishable from zero at this single seed—though this arm should be read with an additional caveat: its own pre-registered manipulation check (target-domain perplexity must drop by at least 0.10) failed, with target-domain perplexity rising slightly rather than falling, so the near-zero DiD here reflects an underpowered manipulation as much as a genuine null. GPT-2’s swap arm passes this check comfortably and is not subject to the same caveat. In both models the placebo shift on a matched random-neuron set is small (0.015 and 0.016 respectively, both below the SESOI = 0.05 threshold), so the metric is not simply reacting to fine-tuning per se, and the canonical entropy neurons mostly survive continued training under the causal-selection gates on the native-domain arm of each model (5 of 6 for GPT-2, 4 of 5 for GPT-Neo-125M), though survival is lower on GPT-Neo’s swapped arm (3 of 5). On GPT-2’s native-domain (OWT) arm—isolating the generic-training effect from any domain switch—carrying *decays with continued training even absent a domain switch*: the entropy-neuron gap (measured under this section’s own pool and split, not Table 1’s—so the pre-training value is not directly comparable to Table 1’s GPT-2 gap of 0.410) falls from 0.341 before training to 0.261, 0.246, and 0.237 at successive token-budget checkpoints, a decay concentrated in the first third of training and tapering thereafter, not a steady linear decline. Since the DiD above (-0.034) is roughly a third of this native-arm decay (-0.105), most of the observed shift under continued training on GPT-2 is attributable to training itself rather than to which domain the model is trained on. We report this decay pattern for GPT-2 specifically: GPT-Neo-125M’s own native arm starts from a gap near zero (it is not an entropy-neuron carrier to begin with, Table 1), so there is too little baseline carrying for a comparable decay to be observable, and we do not extrapolate the decay finding to models that do not carry.

This does not resolve the domain confound of Section 4.4: continued pretraining measures whether carrying *decays* under further training, which is a different quantity from asking whether carrying *originates* from a domain at pretraining time, and a null or noisy decay signal does not license a conclusion about origin. It does, however, provide evidence that lowers the expected value of resolving the confound via full domain-controlled pretraining (Section 6): if a large, deliberate domain manipulation under continued training produces only a small, generic-training-confounded effect, a from-scratch pretrain aimed at isolating a pure data effect is less likely to yield a large, cleanly separable signal than a symmetric a priori expectation would suggest. We report this as Bayesian evidence against the domain-alone hypothesis, not as a formal test of it, and revise our future-work priority accordingly (Section 6).

5 Limitations

The principal limitation is the breadth of the carrying set. The four carriers are the GPT-2 family, one a distillation of another, and they share an architecture, the WebText corpus, a tokenizer, and learned positional encodings; these confounds cannot be separated on pretrained checkpoints, and we make no inferential claim about group means. Several axes that might otherwise explain the split are already excluded by counterexamples within the set: size, training volume, and the tokenizer are each decoupled from carrying, since DistilGPT-2 (120M) carries while OPT-125M and GPT-Neo-125M² do not, GPT-2-large (838M) carries while BLOOM (818M) does not, and Pythia-1.4B, with the most training data here, does not carry. What remains confounded along the lineage is the GPT-2 architecture together with the WebText data and the evaluation-domain match discussed next.

That evaluation-domain match is the most consequential open confound. Our corpus is Pile text [9], in-distribution for Pythia and GPT-Neo and out-of-distribution for GPT-2, so carrying could in part reflect out-of-distribution evaluation. BLOOM and Qwen, also out-of-distribution yet non-carriers, partially undercut this explanation. We have since run the per-domain control this section calls for (Section 4.9): continued pretraining toward the Pile domain produces only a small, generic-training-confounded shift in carrying, and the swapped GPT-Neo arm fails its own pre-registered manipulation check, so the control does not cleanly separate a domain effect from a training-dose effect. This narrows, but does not close, the confound: it bears on whether carrying *decays* under domain-shifted continued training, not on whether it *originates* from a pretraining domain, which only a controlled from-scratch pretrain can settle.

The GPT-Neo-125M/1.3B pair identified in Section 4.7 independently narrows the architecture side of this confound. The two checkpoints are identical in architecture, tokenizer, and pretraining corpus, and differ in whether a distinct causal entropy carrier is present; no single architectural axis we examined (Section 4.8)—positional-encoding scheme, embedding tying, local/global attention pattern—separates carriers from non-carriers cleanly across the full panel, and the two axes with the cleanest one-directional signal (non-learned positional encoding and untied embeddings each predicting *absence* of a carrier, 0 of 4 and 0 of 3 respectively) do not explain why the 125M/1.3B pair, identical on both, differs. This is consistent with a scale- or capacity-dependent mechanism rather than a purely architectural or purely data-driven one, though we did not test scale as an isolated factor and this remains a hypothesis, not a finding.

The analysis also has researcher degrees of freedom. The thresholds in Eqs. 2–4 (the 98th-percentile null gate, $|\Delta H| > 0.02$, norm ≥ 0.30 , the 0.90 activation quantile, $\lambda = 1$, four skipped

²163.8M actual parameters; we use the name GPT-Neo-125M throughout, matching Section 4.7 and its source checkpoint.

tokens, a context of 256) were fixed a priori rather than tuned on the verdict, and we did not sweep them; the verdict is reported under a floor-sensitivity check (Appendix B), but a full hyperparameter sweep remains future work.

One intended covariate proved uninformative on its own terms: the per-model vocabulary hash differs within the GPT-2 family under an identical tokenizer, so it does not certify tokenizer identity, and the tokenizer conclusion rests instead on vocabulary size and the known tokenizer. Our original in-house embedding-tying detector was separately found to be faulty (it returned “untied” for every model, including GPT-2, which we know to tie); Section 4.8 replaces it with tying status verified against each architecture’s released configuration, which is why that comparison rather than an in-house measurement is used above. The question of scale is similarly unresolved: entropy neurons are absent in fully-trained Pythia-160M and TinyStories-33M but present in DistilGPT-2 (120M) and in GPT-Neo and OPT near 125–164M, so a controlled pretrain at ~ 125 M is likely within the trainable range. Without a set of from-scratch-trained models we state controlled pretraining as future work, though Section 4.9’s continued-pretraining evidence lowers our expectation that a domain manipulation alone would yield a large, cleanly separable effect; we do not treat controlled pretraining as uninformative, but we now weight it below the scale-focused variant that Section 4.8 motivates.

A few smaller caveats remain. The evaluation uses a single domain. Cross-model causal magnitudes are not normalized and are therefore not directly comparable. Mean-ablation is inflated by the LayerNorm feedback loop and is reported only relatively. The BLOOM result rests on a single neuron, and the causal sign may reflect a polarity convention. The correlation control selected on test labels in a prior version and is now restricted to the training fold (Appendix B). The R^2_{full} point estimate, finally, lies outside its bootstrap interval in 7 of 11 models. This arises from bootstrap bias: resampling training documents with replacement reduces the number of unique documents and deflates the resampled R^2 . The effect is larger for the high-dimensional full-residual probe than for the low-dimensional entropy-neuron probe, whose point estimates remain inside their intervals, so the carrier verdict is unaffected.

6 Conclusion

Canonical entropy neurons exert a causal influence on output entropy but, outside a single model lineage, do not carry its variation, and the direction of their causal effect is family-dependent. A shared mechanistic signature does not establish a shared functional role. Where output entropy is carried by a distinct causal direction outside that lineage, it is disjoint from the canonical neurons and, in the one architecture where we find it, does not survive scaling within its own family. Both findings dissociate what a representation *predicts* from what it *carries* and what intervening on it *does*, and the pattern is not explained by any single architectural axis we tested, including the pretraining corpus.

Three consequences follow for practice. First, interpretability findings about these neurons do not transfer: steering them to control predictive uncertainty should not be expected to work outside the GPT-2 lineage, where they do not carry the variation, and even where an alternative carrier exists (Section 4.7) it need not persist at a larger scale of the same architecture. Second, “a causal handle is not a carrier” is a general caution—a claim that a neuron or direction implements a function warrants an intervention test, a carrier test, and, where the direction is chosen for its predictive power, an interchange test that a matched unsupervised subspace does not already pass. Third, a decodable representation of a quantity is not evidence that the representation is what the model acts on, a pattern also observed independently in a different setting by [8]; where possible,

causal-necessity and interchange-transfer tests against matched controls should accompany decoding claims.

The three directions posed by an earlier version of this work have each received partial evidence. A per-domain control (Section 4.9) finds that continued pretraining toward the evaluation domain produces only a small, generic-training-confounded shift in carrying, evidence against a clean domain-alone account but not a substitute for a from-scratch test of origin. Identifying a distributed direction that carries entropy outside the canonical neurons (Section 4.7) succeeded in one architecture and, together with the architectural comparison of Section 4.8, redirects the remaining open question from *which architecture or corpus* toward *what changes with scale* within an architecture that already carries at one size. A controlled pretrain remains the only clean test of a data-versus-architecture origin, but we now weight a scale-controlled pretrain within the GPT-Neo architecture—holding architecture and corpus fixed and varying capacity—above a data-controlled pretrain at fixed scale, since our evidence bears more directly on the former.

Ethics and Broader Impact

This is primarily an interpretability study on open-weight language models and public text; it raises no direct ethical concerns. Section 4.9 additionally continues pretraining two small (125M) open-weight models on public text for a few GPU-hours each; this does not create new capabilities or release new weights and raises no concerns beyond the compute reported in the Compute statement below. Better localization of uncertainty-related components may eventually aid calibration and hallucination mitigation; our result is a caution against over-attributing such a role to a single neuron class.

Reproducibility Statement

All gates and hyperparameters are in Eqs. 1–6 and Table 3; seeds, the data hash, and per-model status are recorded with the released artifacts. The paired-bootstrap gap and the deprecated marginal-difference column are released together for an explicit before-and-after audit.

Compute

All experiments ran on a single Tesla T4 GPU, no distributed training. The original 13-model dissociation protocol (Table 1, Figures 1–5) required ≈ 4.5 fp32 GPU-hours. The interchange-carrier panel (Table 2, Figure 6: four separate acceptance-verified runs, 42 (model, seed) combinations in total) required ≈ 10 further GPU-hours, and the continued-pretraining pilot of Section 4.9 required ≈ 10 further GPU-hours across its four completed arms (excluding smoke tests). Total across all reported experiments is ≈ 25 GPU-hours; exact per-run times are recorded in the released `run_meta_*.json` files.

Data and Artifact Availability

Code, frozen result tables, and run metadata (including the data hash) are released; models and data are public (TransformerLens / HuggingFace; NeelNanda/pile-10k).

A Synthetic validation

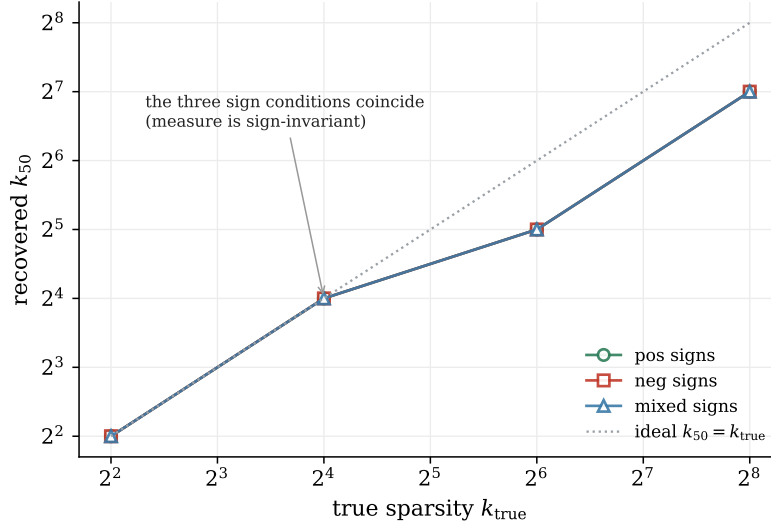


Figure 7: Synthetic validation of the sign-invariant attribution measure. Recovered sparsity k_{50} against the planted sparsity k_{true} for positive, negative, and mixed contribution signs; the three coincide (the measure is sign-invariant) and track the ideal up to the resolution limit. Recall of the planted set is 1.0 throughout.

Full protocol and the recall, sparsity-recovery and sign-invariance results summarized in Section 3.3 and Figure 7. The measure validates recall and sign-invariance, not precision.

B Reproducibility details

The thirteen models span seven families: GPT-2 and DistilGPT-2 [18], the Pythia suite [2], GPT-Neo [4], OPT [21], BLOOM [3], Qwen2.5 [17], and TinyStories [6]. Evaluation uses Pile text [9]. The selection and probing methodology follows the sparse-probing and entropy-neuron literature [11, 19]. Table 3 lists the full configuration and environment; Tables 4–6 give the activation- and entropy-variance descriptors, the deprecated-versus-corrected gap, and the criterion-overlap summary referenced in the main text.

C The “why GPT-2” battery

The geometry probe (with the passed-null-gate control), the scale-invariant ablation-channel measure, and the causal-direction injection test, all reported descriptively (Figure 8 and Section 4.6), with the circularity, the instability of the causal-direction statistic, and the broken tying and vocabulary-hash covariates disclosed.

Setting	Value
Eval documents per model	60
Selection/eval pools	disjoint halves of 140 Pile documents
Max tokens / skipped tokens	256 / 4
High-activation quantile	0.90
Null-space percentile gate	98
$ \Delta H $, norm, flip gates	> 0.02 nats, ≥ 0.30 , < 0.01
Max entropy neurons	6
Ridge λ	1.0
Train fraction	0.70
Bootstrap replicates	1000
Causal documents / direction-test documents	≤ 12 / 8
Seeds	{0, 1, 2}
Precision / GPU	fp32 / Tesla T4
Data hash	da5f6242b06fa229
Library versions	torch 2.7.1, transformer_lens 3.4.0, transformers 5.10.2, datasets 4.8.5, pandas 2.3.3

Table 3: Hyperparameters and environment for full reproduction. Verdicts are additionally reported under floor-sensitivity at R_{EN}^2 thresholds {0.05, 0.10, 0.15}; the four carriers survive all three.

Model	EN act. variance	entropy variance
GPT-2	8.64	4.09
GPT-2-medium	2.14	3.99
GPT-2-large	0.30	3.43
DistilGPT-2	0.46	3.91
Pythia-410M	15.87	3.17
Pythia-1B	0.78	3.01
Pythia-1.4B	1.32	2.90
GPT-Neo-125M	1.30	3.74
BLOOM-560M	1.45	3.53
Qwen2.5-0.5B	50.59	3.10
OPT-125M	0.00	3.79

Table 4: Activation- and entropy-variance descriptors (Section 4.6). Carrying is, if anything, inversely related to entropy-neuron activation variance: the two models with the largest variance (Qwen, a fragile near-miss, and Pythia-410M, a non-carrier) do not robustly carry, while DistilGPT-2, with small variance, is among the strongest carriers. Carrier entropy variance (3.43–4.09) overlaps the non-carrier range (2.90–3.79).

Model	deprecated gap_{lo}	paired gap_{lo}
GPT-2	0.157	0.197
GPT-2-medium	0.141	0.191
GPT-2-large	0.074	0.116
DistilGPT-2	0.222	0.279
Pythia-410M	-0.167	-0.132
Pythia-1B	-0.085	-0.066
Pythia-1.4B	-0.079	-0.037
GPT-Neo-125M	-0.175	-0.157
BLOOM-560M	-0.054	-0.046
Qwen2.5-0.5B	-0.077	0.010
OPT-125M	-0.111	-0.100

Table 5: The deprecated marginal-difference gap versus the corrected paired-bootstrap gap (Eq. 5). The correction widens every margin because it removes the spurious inflation from treating two positively correlated quantities as independent; the four carrier verdicts are unchanged, and Qwen moves from a negative deprecated bound to a small positive paired bound, consistent with its near-miss status.

Model	top-5 overlap
GPT-2-large	1
DistilGPT-2	1
all other nine models	0

Table 6: Overlap between the top-five entropy-neuron sets selected by the null-space and correlation-with-entropy criteria (out of five). The two criteria share at most one neuron in any model (indices 4537 in GPT-2-large, 1611 in DistilGPT-2) and none in the other nine, so they identify different units (Section 4.5). The full top-five index lists are released with the artifacts.

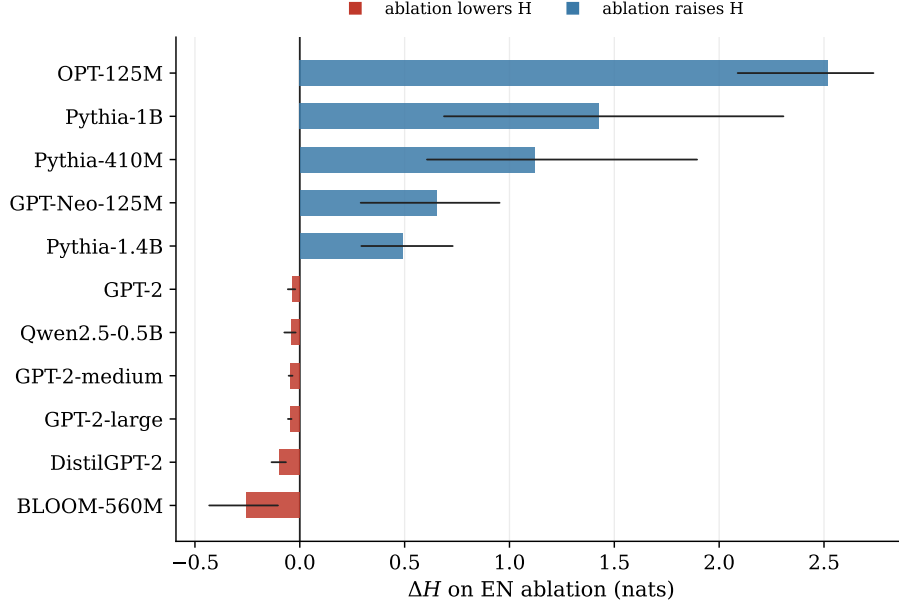


Figure 8: Family-dependent sign of the causal effect. ΔH under entropy-neuron ablation per model, with document-level intervals; ablation lowers entropy in the GPT-2 lineage, BLOOM and Qwen, and raises it elsewhere. Reported descriptively: the sign does not track carrying. Wide intervals for Pythia and OPT reflect the small causal-document count (≤ 12).

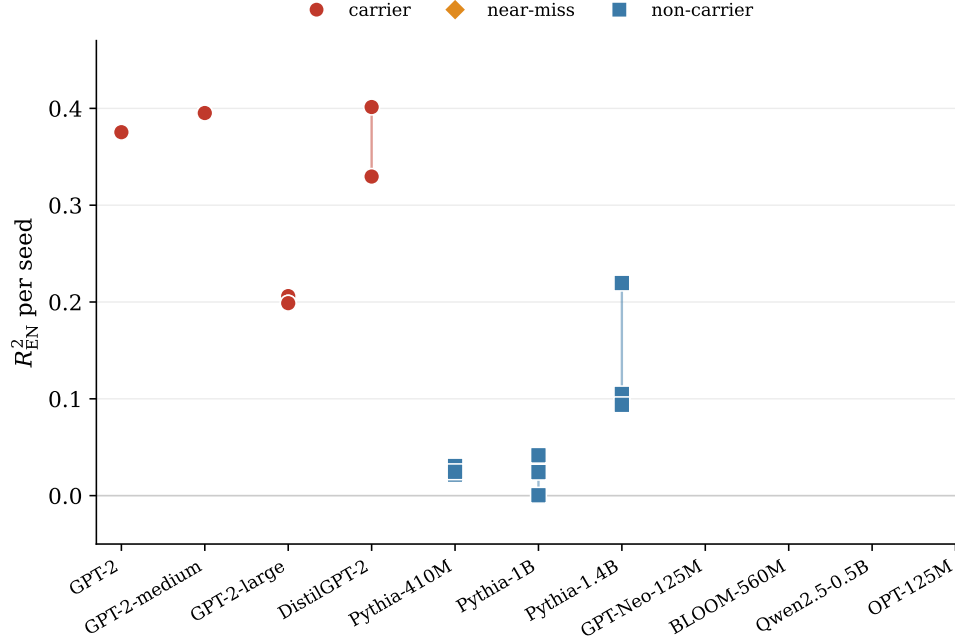


Figure 9: Per-seed R^2_{EN} under the three selection seeds, in canonical model order. Carrier estimates are high and tightly clustered; the two near-misses are dispersed, with Qwen collapsing to 0.067 on its worst seed, which is what the worst-seed rule (Eq. 6) acts on. Non-carriers remain near zero. EN: entropy neurons.

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